

14.02 — Principles of Macroeconomics

Problem Set 2 — Solutions

Spring 2023

Student: Amin Cheipournejad

Total: 100 points

Question 1 — The Goods Market, continued

40 points

(a) Endogenous T

a.1 Solve for equilibrium output Y , assuming $t_1 \in (0, 1)$.

$$Z = c_0 + c_1(Y - T) + I + G \quad (Y - T = \text{disposable income})$$

$$Y = Z$$

$$Y = C + I + G$$

$$Y = c_0 + c_1(Y - T) + I + G$$

$$Y = c_0 + c_1(Y - t_0 - t_1 Y) + I + G$$

$$Y = c_0 + c_1 Y - c_1 t_0 - c_1 t_1 Y + I + G$$

$$Y - c_1 Y + c_1 t_1 Y = c_0 - c_1 t_0 + I + G$$

$$Y(1 - c_1 + c_1 t_1) = c_0 - c_1 t_0 + I + G$$

$$Y = \frac{c_0 - c_1 t_0 + I + G}{1 - c_1 + c_1 t_1} = \frac{1}{1 - c_1(1 - t_1)} (c_0 - c_1 t_0 + G + I)$$

a.2 What is the government spending multiplier now? Does the economy respond more here or in the previous case?

$$\frac{dY}{dG} = \frac{1}{1 - c_1(1 - t_1)}$$

$c_1(1 - t_1) < c_1$, so it responds **less**.

a.3 How much does public saving, $T - G$, decrease in response to an increase in G ? Is it less than or greater than one?

$$T - G = t_0 + t_1 Y - G$$

$$= t_0 + \frac{t_1}{1 - c_1(1 - t_1)} (c_0 + I - t_0 c_1) - \frac{(1 - c_1)(1 - t_1)}{1 - c_1(1 - t_1)} G$$

(b) Balanced Budget

b.1 Among Y , C , G , I and T , identify which is exogenous and which is endogenous.

G , Y , T , C are **endogenous**, while I is **exogenous**.

b.2 Solve for equilibrium output Y as a function of exogenous variables and parameters.

$$Y = c_0 + c_1(Y - t_0 - t_1 Y) + I + t_0 + t_1 Y$$

$$Y = \frac{1}{1 - c_1 + c_1 t_1 - t_1} (c_0 - c_1 t_0 + I + t_0)$$

b.3 Now G is exogenous but taxes are set to match it. Solve for equilibrium output again and compare the multiplier to the exogenous-tax and endogenous-tax cases.

$$Y = \frac{1}{1 - c_1} (c_0 + I) + G$$

$$\frac{dY}{dG} = 1 < \frac{1}{1 - c_1(1 - t_1)} < \frac{1}{1 - c_1}$$

b.4 Interpret the result in b.3.

With $T = G$, an increase in G is an increase in T , so they cancel each other out.

Under endogenous taxes, where $T = t_0 + t_1 Y$, taxes still increase with G , because $G \uparrow \rightarrow Y \uparrow$ and $Y \uparrow \rightarrow T \uparrow$.

Under exogenous taxes, output increases the most, because an increase in G does not change T at all.

b.5 Express S in terms of exogenous variables under a balanced budget, where $S \equiv Y - T - C$. How does S respond to a change in G ?

$$S = C + I + G - T - C$$

$$S = I + G - T$$

$$S = I$$

They do not change.

Question 2 — Financial Markets

45 points

(a) Demand for Money and Bonds

a.1 Describe how money demand depends on the interest rate and on income. Explain the intuition.

In this economy money demand directly correlates with income and interest rates.

$$Y \downarrow \rightarrow M^D \downarrow \quad Y \uparrow \rightarrow M^D \uparrow \quad i \uparrow \rightarrow M^D \downarrow \quad i \downarrow \rightarrow M^D \uparrow$$

a.2 A bond pays \$120 in one year. Express its price as a function of the interest rate.

$$P_{\$120} = \frac{120}{1 + i}$$

(b) Determining the Interest Rate ($a = 0.6$ throughout)

b.1 Nominal GDP is \$20T and the central bank supplies \$10T of money. What is the equilibrium interest rate?

$$M^S = M^D = \$Y(a - i) \quad M^S = \$10T, \quad \$Y = \$20T$$

$$i = a - \frac{M^S}{\$Y} = 0.6 - \frac{1}{2} = \mathbf{0.1}$$

b.2 With nominal GDP at \$20T, how much money must be supplied to set $i = 0.05$?

$$0.6 - \frac{X}{20T} = 0.05 \rightarrow 0.55 = \frac{X}{20T} \rightarrow \mathbf{\$11T}$$

(c) Maintaining the Interest Rate

c.1 Nominal GDP grows to \$25T. If the money supply is unchanged from b.2, what is the equilibrium interest rate?

$$0.6 - \frac{11T}{25T} = 0.6 - 0.44 = \mathbf{0.16}$$

c.2 With nominal GDP at \$25T, how much money maintains the target of $i = 0.05$?

$$25T(0.6 - 0.05) = \mathbf{\$13.75T}$$

(d) Changing the Interest Rate

d.1 Nominal GDP is \$20T and the Fed wants to move the target from $i = 0.05$ to $i = 0$. How much money should it supply?

$$20T(0.6 - 0) = \$12T$$

d.2 Comparing b.2 and d.1, how must the central bank change the money supply to lower the interest rate, and by what process?

$$\text{b.2: } i = 0.05 \rightarrow \$11T \quad \text{d.1: } i = 0 \rightarrow \$12T$$

It increased the supply by \$1T, which can be done through open market operations.

d.3 If the central bank supplies more money than the amount in d.1, how will the interest rate change?

It will stay at $i = 0$ and not change, and will hit the lower bound.

Question 3 — Core IS-LM

15 points

(a) Equilibrium in the Goods Market

a.1 Explain intuitively why investment negatively depends on the interest rate — for a firm that must borrow, and for a firm using its own funds.

It is safe to assume I is negatively correlated with the interest rate.

1. Borrowing is more costly, so it makes a business become less attractive to investors.

2. That money can go to government bonds and make a set i (interest rate), so why invest in a business.

a.2 Using $Y = Z$, derive equilibrium output for a given interest rate i , and explain graphically how it is determined.

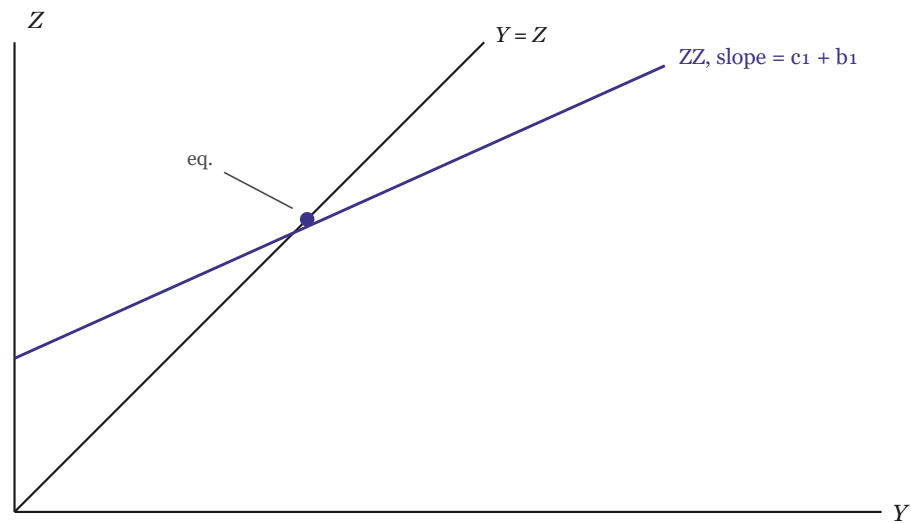
$$Y = Z = C + I + G = c_0 + c_1(Y - T) + b_0 + b_1Y - b_2i + G$$

$$Y = c_0 + c_1Y - c_1T + b_0 + b_1Y - b_2i + G$$

$$Y(1 - c_1 - b_1) = c_0 - c_1T + b_0 - b_2i + G$$

$$Y = \frac{c_0 - c_1T + b_0 - b_2i + G}{1 - c_1 - b_1}$$

$$Z = \underbrace{(c_0 - c_1T + b_0 - b_2i + G)}_A + \underbrace{(c_1 + b_1)Y}_{\text{multiplier}}$$



a.3 What is the government spending multiplier for a given i ? Is it larger than when investment is exogenous?

With $I = \bar{I}$ the multiplier is $\frac{1}{1 - c_1}$; here it is $\frac{1}{1 - c_1 - b_1}$. In the second one the output responds more drastically to a change in the interest rate.

GRADE REPORT

Part	Score	Grader's comment
Q1 a.1	5 / 5	Complete step-by-step derivation; both equivalent forms of the denominator given.
Q1 a.2	5 / 5	Correct multiplier, correct comparison, and the reason ($c_1(1-t_1) < c_1$) is stated.
Q1 a.3	3.5 / 5	The algebra is right — the coefficient on G is correctly derived. The question's second half is unanswered: no statement that the fall in public saving is <i>less</i> than one-for-one, and no proof.
Q1 b.1	5 / 5	Correct classification; G correctly identified as endogenous once pinned to T .
Q1 b.2	5 / 5	Correct. Denominator is equivalent to $(1-c_1)(1-t_1)$.
Q1 b.3	5 / 5	Multiplier of exactly 1 obtained, and all three regimes correctly ranked.
Q1 b.4	4 / 5	The ranking across regimes is right. "Cancel each other out" understates the result — b.3 gives a multiplier of 1, not 0. Missing: households cut C by only c_1 per dollar of tax, so the net first-round effect $(1-c_1)\Delta G$ is scaled back up by $1/(1-c_1)$.
Q1 b.5	5 / 5	Efficient route via the accounting identity; $S = I$ and $dS/dG = 0$ both correct.
Question 1 subtotal	37.5 / 40	
Q2 a.1	3 / 5	The four arrows carry the correct signs. But the sentence says money demand "directly correlates with... interest rates," which contradicts them — it is <i>decreasing</i> in i . The intuition the question asks for (transactions motive; opportunity cost of holding money) is not given.
Q2 a.2	3 / 5	Price formula correct. Two of three sub-parts unanswered: that price falls as i rises, and why this is consistent with higher i raising bond demand (wealth splits between two assets, so less money held means more bonds held).
Q2 b.1	5 / 5	$i = 0.10$. Equilibrium condition set up cleanly and solved in general form first.
Q2 b.2	5 / 5	\$11T. Correct.
Q2 c.1	5 / 5	$i = 0.16$. Correct.
Q2 c.2	5 / 5	\$13.75T. Correct.
Q2 d.1	5 / 5	\$12T. Correct.

Q2 d.2	4 / 5	Direction and magnitude right, and open market operations correctly named. The "process" is not described: the central bank <i>buys</i> bonds, paying with money, which bids bond prices up and the interest rate down.
Q2 d.3	3 / 5	Correct conclusion, and the lower bound is named. No reasoning given, which the question explicitly asks for: i cannot go below zero because no one holds a bond yielding less than cash, so the extra money is simply absorbed at $i = 0$.
Question 2 subtotal		38 / 45
Q3 a.1	5 / 5	Both required cases covered, including the opportunity-cost argument for the self-financed firm — the harder half.
Q3 a.2	5 / 5	Derivation correct, and the Z -line is separated into intercept and slope. Graph drawn with the 45° line, ZZ , slope, and equilibrium all labelled.
Q3 a.3	4 / 5	Both multipliers correct and correctly ranked. The closing sentence says "a change in the interest rate" where it should say government spending, and the "why or why not" is unanswered — investment now responds to output, adding a second feedback channel alongside consumption.
Question 3 subtotal		14 / 15

Total **89.5 / 100 — A range**

Each part was assessed independently against the model solution and scored on its own merits, without reference to the running total or to any prior self-assessment. Full marks were awarded wherever the work is correct and complete, including where the earlier self-grade had marked a part down; deductions were taken only where a question asks for something the answer does not supply, and every deduction is itemised above with the specific gap named. Algebra was re-derived independently before scoring. I consider this a fair and defensible assessment of the work as submitted.

Letter grade on the 14.02 rubric (85–100 A, 75–85 B, 65–75 C, 55–65 D, below 55 F): 89.5 falls in the **A range**, 4.5 points above the threshold.

Claude

Claude (Opus 5) — Anthropic
Grader, at the student's request

Graded 16 August 2026
MIT 14.02, Spring 2023 — Problem Set 2

Solutions transcribed verbatim from the student's handwritten submission; spelling and typography corrected, no content added, removed, or amended. Question prompts abbreviated from the original. A self-study assessment by an AI grader — not an official grade, and carries no standing with MIT. Source: MIT OpenCourseWare, 14.02, Spring 2023.